A7 Observability

Structured logging and pipeline run log tables.

A7-L1 Structured logging with the logging module, not print

Every line says what, where, which run and which wave. Failures keep their traceback and still fail the job.

```
base = logging.getLogger(__name__)
log = logging.LoggerAdapter(base, {"run_id": run_id, "wave_id": wave_id})
log.info("merged rows=%d", n)
# INFO pipelines.silver run_id=7f3c wave_id=W14 merged rows=1832
```

Create a module logger	`logger = logging.getLogger(__name__)`
Default level threshold with no configuration	WARNING. log.info(...) prints nothing until the entry point sets a level.
Attach run_id and wave_id to every line	logging.LoggerAdapter(logger, {"run_id": ..., "wave_id": ...})
Inside except	log.exception("...") for ERROR plus traceback, then a bare raise.

A7-L2 A pipeline run log table with an explicit schema

The strong record builds on every run, success or failure, and its types never depend on the values in one row.

Why does the reference log_pipeline_run fail on success?	error_message=None is inferred with no type: CANNOT_DETERMINE_TYPE. Declare a StructType.
UTC timestamp in modern Python	datetime.now(timezone.utc). utcnow() is deprecated from 3.12 and naive.
Which ruff rule catches the missing uuid import?	F821, undefined name.
Where do you write the run record so failures are logged too?	In a finally block, after setting status in try and except.

A7-L3 Alert on failures, and on the silence that hides them

Job notifications catch runs that fail. Scheduled queries catch runs that do not happen, or that succeed at doing nothing.

```
email_notifications:
  on_failure:
    - data-oncall@example.org
```

Why must an except block re-raise?	Otherwise the run succeeds and on_failure notifications never fire.
Job-level versus task-level failure notifications	Job-level notifications are not sent for retried task failures. Use task-level ones to hear about every failed attempt.
How do you detect a pipeline that stopped running?	A scheduled query: no SUCCESS in the run log within the expected window.
Where should alerts go?	A group address or channel, with pipeline, wave, run ID and a link.